

WID2001
KNOWLEDGE REPRESENTATION AND
REASONING

Semester 2, Session 2025/2026

Group Project (20%)

You are required to work in small groups (five (5) to six (6) people) that include domain experts (DE), knowledge engineer (KE), programmer (Pr), expected end user (EU), and a project manager (PM). One of the roles can have more than one member except PM.

Your task is to develop an expert system in a domain of your choice. The prototype must include at least three core components: a knowledge base (KB), an inference engine (IE), and a user interface (UI). Additional modules, such as uncertainty handling and explanation facilities, are optional.

First, the team must communicate with and obtain approval from a selected real human expert (e.g., a professor/industry expert). The knowledge engineer will then apply knowledge elicitation and acquisition techniques to extract expertise related to the chosen domain. Meanwhile, the domain expert should deepen and expand their understanding of the subject by consulting reliable references. Both the knowledge engineer and the domain expert must also meet with the end user to clarify requirements.

All team members will then collaborate with the programmer to design and develop the system, using an appropriate intermediate knowledge representation language and selecting a suitable inference method. The prototype can be built either from scratch or by using existing expert system shells or tools for rapid development.

Based on a chosen domain aligned with a relevant Sustainable Development Goal (SDG), and informed by each member's review of expert system tutorials, assign roles to each team member. Each group must then develop a prototype of the proposed expert system by completing the following tasks:

Task 1: A proposal slide report with the following contents: (10%) – Submit by **Week 11 (Monday)**

1. Introduction
2. Objective
3. Literature Review: Related and recent available ES
4. Specific Domain (Unique use-case)
5. Subject Matter Expert from the field/ Industry (to support the knowledge of the 'DE')
6. ES Architecture (design)- method & solution (at least with the 3 components): KB, IE, UI
7. Implementation –Selection & Mastery of the tool: ES Software & Tools
8. ES project planner (Gantt chart & roles)
9. Testing (Human Expert & End-users involvement)
10. Discussion / results
11. Future and Conclusion
12. Report Format (table/figures/ Gantt chart, submission dateline) and References

Task 2: A Group Presentation and a Demonstration of the ES Prototype (recorded video) (3-4 minutes) (10%) – **Week 13 (Monday)**

1. All group members MUST be available and please switch on your camera during the presentation.
2. You need to present and do demonstration of your ES prototype within the allocated time slots.

SDG reference: <https://sdgs.un.org/goals>